

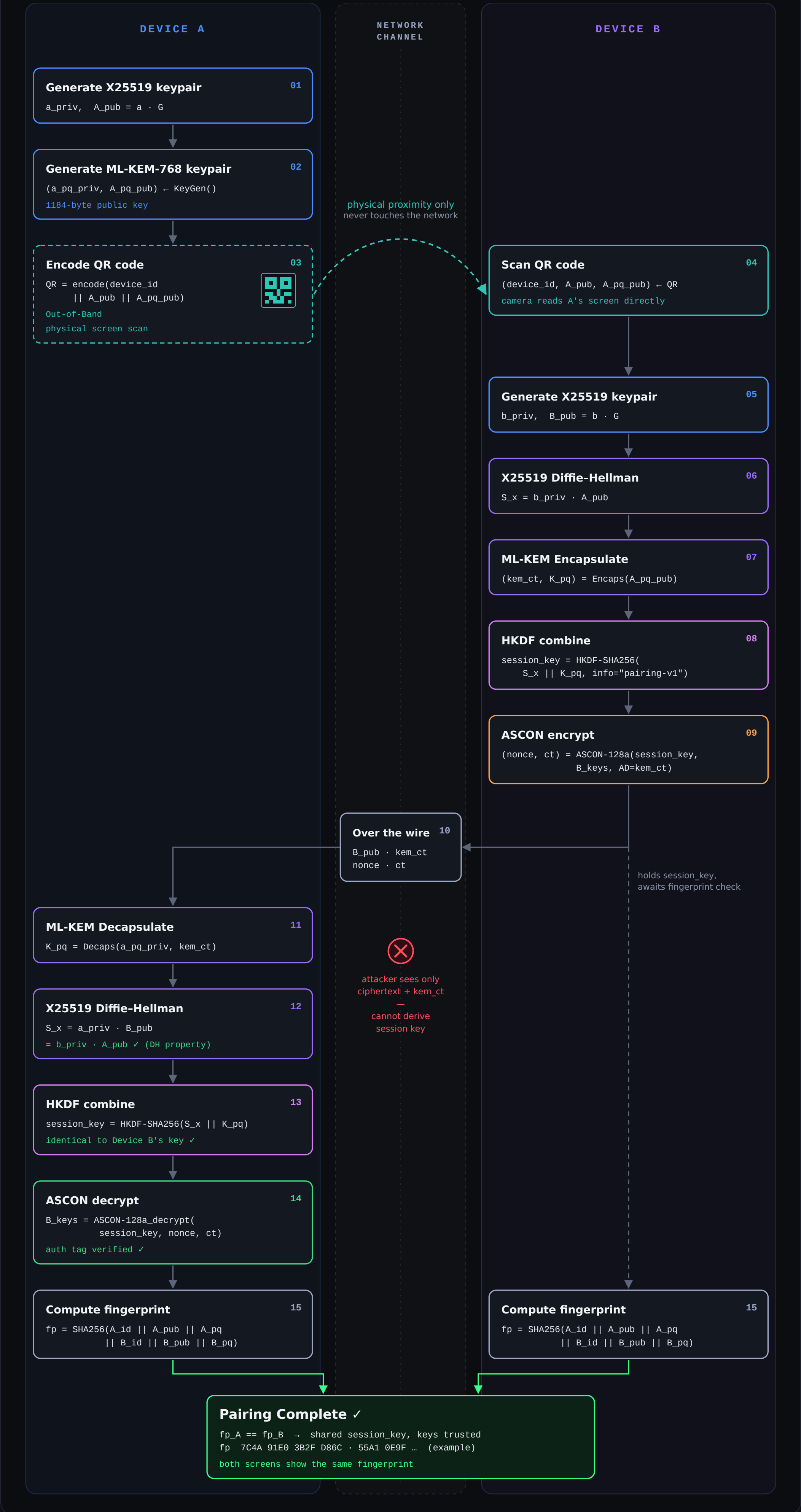
Hybrid post-quantum device pairing

How two devices agree on a session key through a QR scan and one network message, and why a man-in-the-middle can't get between them.

1 · Secure pairing protocol

Device A publishes its public keys out-of-band via QR. Device B derives a hybrid session key and sends back one encrypted message. The session key never crosses the network.

Key generation DH / KEM HKDF Encrypt Decrypt QR / out-of-band Success



B_{pub} travels in the clear next to kem_ct . Device A needs it to compute S_x before it can decrypt ct . That's safe: a public key reveals nothing without a_{priv} .

2 · MITM attack & why it fails

Eve tries to swap in her own keys and relay traffic between the devices. Two independent defences stop her: the out-of-band QR scan and the fingerprint comparison.

